# Beyond Retrieval: Scope-Aware Memory Application in AI Agents

Isaac Chin Wei En
A0248885H

Edward Chew Kian Boon
A0271710N

Ng Cheng Kiat Nicholas
A0271938R

Jagan Mohan Shailesh
A0272542H

**Abstract**

Long-term memory enables language-model agents to retain user preferences across interactions, but retrieving a relevant memory does not guarantee that the agent applies it in the correct context. We investigate scope-aware memory application by augmenting Mem0 and Supermemory with write-time and read-time scope tagging. We construct a benchmark of 40 episodes in 20 paired scenarios, covering five types of conflicting instructions, and evaluate eight conditions across 320 runs. The expected memory was retrieved in 43 of 51 incorrect responses across the six memory-system conditions, demonstrating a substantial retrieval–application gap. Read-time tagging with Supermemory achieves the highest observed adherence (92.5%), compared with 85% for its baseline, although this difference is not statistically significant ($p = 0.25$). Write-time tagging scored lower than the baseline for both systems (not significantly: $p = 0.118$ and $p = 0.549$); for Mem0 it also scored lower than read-time tagging (60% vs 80%, $p = 0.039$), which is consistent with Mem0 re-extracting the pre-tagged instructions. These findings suggest that explicit scope information may support memory application, but its effectiveness depends on how it is integrated and used.

## 1 Introduction

### 1.1 Motivation and Research Problem

Memory-augmented language-model agents retain user instructions across conversations, allowing responses to reflect previously expressed preferences. However, preferences are not always universally applicable. A user may generally prefer Polars for large datasets while requiring pandas for a particular group project. Both memories remain valid, but the appropriate action depends on the current context.

Existing memory systems primarily address the storage and retrieval of relevant information. Yet even when conflicting instructions are successfully retrieved, an agent must determine which instruction applies, whether an exception overrides a standing preference, and whether a temporary instruction should continue to influence subsequent tasks.

We investigate this distinction between *memory retrieval* and *memory application*: does an agent correctly use a retrieved instruction within its intended scope, and can explicit scope information improve its decisions?

### 1.2 Research Objectives and Contributions

We investigate two questions: (1) whether agents make incorrect decisions despite retrieving the expected memory, and (2) whether scope-aware interventions improve contextual adherence

while preserving unrelated preferences.

Our contributions are:

1. A lightweight scope-tagging layer supporting standing, project-specific, and single-task instructions, integrated with Mem0 and Supermemory at write time or read time.
2. A synthetic benchmark of 20 paired scenarios (40 episodes) covering five memory-conflict categories.
3. An evaluation of eight conditions across 320 runs, separating retrieval success, memory application, scope errors, tagging accuracy, and unrelated-memory retention.

# 2 Related Work

## 2.1 Long-Term Memory in LLM Agents

Mem0 [1] provides persistent agent memory through extraction, storage, updating, and retrieval of information from interactions. Supermemory [6] offers a hosted memory service with ingestion and semantic search. Both aim to make relevant prior information available to agents without repeatedly supplying complete conversation histories.

Our work uses these systems as memory backends rather than proposing another storage or retrieval mechanism. We evaluate whether their retrieved information supports contextually appropriate decisions.

Existing long-term memory benchmarks mainly score recall. LongMemEval [8] and LoCoMo [3] test whether an assistant answers questions about earlier conversations (LoCoMo also covers event summarisation and dialogue generation), and ConvoMem [5] compares RAG-based memory systems, including Mem0, with placing the whole history in context. Our benchmark differs in what it scores: not whether a fact can be recalled, but whether the retrieved instruction is applied inside its scope and left unapplied outside it.

## 2.2 Memory Retrieval and Contextual Applicability

Retrieval relevance and contextual applicability are distinct. A memory may be relevant to the current topic but inapplicable to the current project, task, or condition. Conversely, a retrieved exception may need to override a broader preference without permanently replacing it.

We focus on this downstream decision problem by representing scope explicitly and evaluating both correct application and incorrect application beyond a memory's intended context.

# 3 Methodology

## 3.1 Scope Representation and Conflict Resolution

Each instruction is represented by three fields: *scope*, *key*, and *condition*. The scope determines applicability, the optional key identifies a project or task, and the optional condition specifies an additional constraint.

The reference decision policy first excludes memories whose project, task, or condition does not match the current request. Among applicable memories concerning the same decision, scope precedence is

$$\text{single-task} \succ \text{project} \succ \text{standing}.$$

Table 1: Scope representation.

| Scope | Applicability | Example |
|---|---|---|
| Standing | General preference across tasks | Use Polars |
| Project | Restricted to a named project | Marlin: pandas |
| Single-task | Restricted to one task | This export: CSV |

Within one scope, an applicable conditional instruction takes precedence over an unconditional instruction; recency resolves remaining conflicts at equal specificity. Unrelated instructions do not compete, and the benchmark excludes unresolved ties.

These rules define the reference actions and are communicated to the agent. The scope layer does not deterministically select the winning memory.

### 3.2 Scope-Tagging Architecture

We augment Mem0 [1] and Supermemory [6] with an LLM-based scope tagger. Figure 1 compares the two integration strategies.

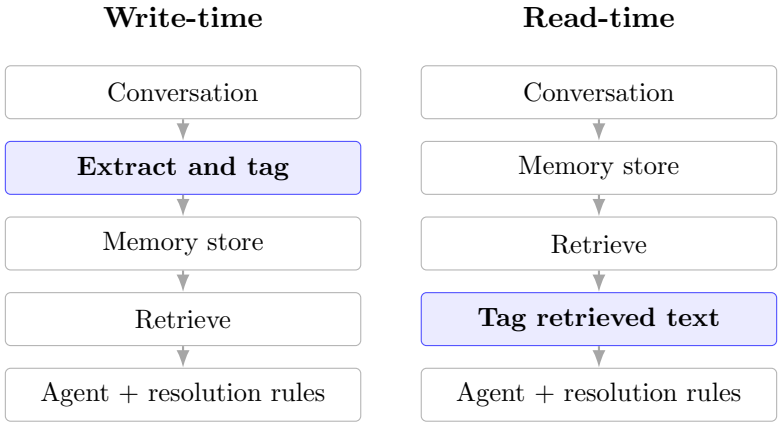



Figure 1: Scope-tagging integration strategies. Write-time tagging occurs before storage, whereas read-time tagging occurs after retrieval.

Write-time tagging extracts individual instructions from the full conversation and stores them with scope metadata. It can preserve scope cues early but may discard surrounding conversational context. Read-time tagging operates on the memory system's retrieved text, preserving native storage behaviour but relying on that text to retain sufficient contextual information.

Both strategies supply tagged memories and conflict-resolution rules to the agent. Structured task identifiers are withheld from the taggers, so task continuity must be inferred from natural-language context.

The two memory systems return different units. Mem0 extracts and rewrites short notes. Supermemory's search returns raw session passages (retrieved items begin "User: ... Assistant: ..."), so its baseline prompt is close to the full transcript in size (444 vs 465 tokens per episode on average). A read-time tag on a Supermemory result therefore labels a whole passage with one scope, even when the passage holds several instructions.

In the write-time variants each tagged instruction is stored with its tag as metadata through the system's normal write call. For Mem0 that call runs its default extraction (`infer=True`), so the pre-extracted instruction is rewritten again before storage.

## 3.3 Benchmark Construction

### 3.3.1 Episode Structure and Mirrored-Pair Design

We construct 20 paired scenarios containing 40 episodes. Each pair includes two or three earlier conversation sessions, reference memories, and two probes sharing the same history. Probe A tests an applicable scoped instruction; Probe B changes the project, task, or condition so that the instruction no longer applies. Each episode consists of one probe and its shared history, with an expected action for evaluation.

Each pair also includes an unrelated standing preference and a retention check. Pair accuracy requires both probes to be correct, discouraging fixed choices that ignore context.

For example, pair `po-01` combines a standing preference for Polars on datasets exceeding one million rows with a project-specific instruction to use pandas for Project Marlin.

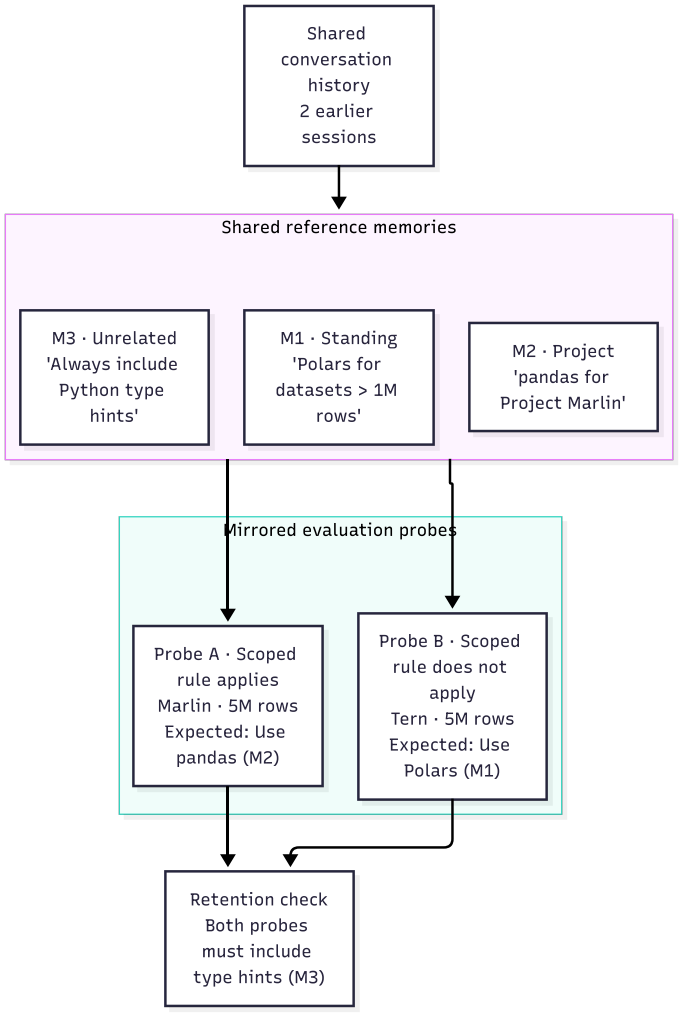



Figure 2: Mirrored-pair benchmark design illustrated using pair `po-01`. Both probes share the same conversation history and reference memories but differ in whether the project-specific instruction applies. An unrelated standing preference is evaluated independently in both probes.

### 3.3.2 Conflict Categories and Dataset Generation

The benchmark covers five conflict categories, with four pairs per category:

The categories test project exceptions, temporary task exceptions, conditional applicability,

Table 2: Benchmark composition.

| Conflict category | Pairs | Episodes |
|---|---|---|
| Project override | 4 | 8 |
| Single-task override | 4 | 8 |
| Conditional scope | 4 | 8 |
| Same-scope supersession | 4 | 8 |
| Same-scope specificity | 4 | 8 |
| Total | 20 | 40 |

replacement of older instructions within one scope, and precedence of more specific instructions within one scope.

The 20 pairs contain 66 distinct reference memories, including competing and unrelated instructions. Nineteen pairs were initially drafted with AI assistance and subsequently reviewed and revised. The suite varies domains, wording, chronology, and instruction positions; some standing preferences deliberately appear after narrower instructions.

#### 3.3.3 Dataset Validation and Gold Labels

Automated checks validate the JSON schema, verbatim memory quotations, paired-probe similarity, expected decisions, and diversity. Only session and probe text reach the systems; structured gold labels and expected actions are withheld. A leakage test checks this separation.

The 66 memories were reviewed against the drafted labels. The reviewed labels matched the drafts on scope and key for all 66 memories and differed on four condition expressions (94% condition agreement with the drafts); the original reference expressions were retained. The two committed annotation files are identical, so we report no inter-annotator agreement.

### 3.4 Experimental Conditions and Baselines

We evaluate eight conditions: no memory, full transcript, Mem0, Supermemory, and write-time/read-time scope-tagged variants of each memory system.

Table 3: Experimental conditions.

| Condition | Memory supplied to agent |
|---|---|
| No memory | Current probe only |
| Full transcript | All earlier sessions |
| Mem0 | Native retrieved memories |
| Supermemory | Native retrieved search results |
| Mem0 + write | Pre-extracted, tagged memories |
| Mem0 + read | Retrieved memories tagged afterward |
| Supermemory + write | Pre-extracted, tagged memories |
| Supermemory + read | Retrieved results tagged afterward |

Each condition evaluates all 40 episodes, yielding 320 runs. The scope-layer conditions introduce tags, resolution rules, and memory ordering information together; their individual effects cannot be isolated.

# 4 Experimental Setup and Evaluation

## 4.1 Implementation and Model Configuration

The implementation uses Python 3.11+, `uv`, and a shared lockfile. The agent is `qwen/qwen3.8-flash` through OpenRouter, with temperature 0, seed 4213, a 1,024-token output limit, and reasoning disabled. The agent prompt template and sampling settings are fixed across conditions, with retrieval limited to ten results.

Qwen was selected after an operational pilot: it reported zero reasoning usage on 40 pilot calls, whereas DeepSeek returned empty text on three of eight calls when reasoning was disabled.

Both memory systems use a common interface and fresh namespaces for each episode and condition. Mem0 uses its default extraction and search, an `all-MiniLM-L6-v2` embedder (384 dimensions), Qdrant storage, and internal LLM calls through OpenRouter. The locked environment omits Mem0's optional spaCy and fastembed dependencies, so all Mem0 runs used vector retrieval without BM25. Supermemory uses its hosted API and waits for ingestion to complete before retrieval.

OpenRouter token usage and costs are recorded in an append-only ledger. Supermemory's separately billed processing is excluded from those costs.

## 4.2 Evaluation Metrics

An LLM judge identifies the action taken in each free-text answer. We evaluate the following metrics:

Table 4: Evaluation metrics.

| Metric | Definition |
|---|---|
| Adherence | Expected action selected |
| Pair accuracy | Both A and B probes correct |
| Retrieval–application gap | Expected memory retrieved, wrong action selected |
| Retention | Unrelated standing preference followed |
| Tag accuracy | Predicted scope metadata matches reference |
| Failure stage | Earliest detected pipeline failure |

Adherence is also separated into in-scope (A) and out-of-scope (B) probes. Incorrect decisions are classified as over-application, under-application, supersession, or other actions.

Retrieval is checked by verbatim matching or judge-assisted matching of rewritten notes. The full-transcript control counts as successful retrieval by construction; no memory counts as unsuccessful. Retention excludes unassessable answers.

Tagging accuracy is measured on notes associated with one reference memory. Scope is matched exactly; project keys must contain the reference project name, while single-task keys are excluded from scoring. Conditions use strict (variable, operator, value) and lenient (operator, value) matching.

Failure-stage categories are extraction, retrieval, scope loss, tagging, and reasoning. Each incorrect answer is assigned to the earliest detected failure. These are diagnostic classifications, not established causal mechanisms. Supermemory's internal extraction cannot be inspected separately from retrieval.

### 4.3 LLM-as-a-Judge and Statistical Analysis

The judge uses the same Qwen model at temperature 0. It receives the probe, response, and alphabetically ordered candidate actions, but not the expected action. It returns an action, `none`, or `mixed`, with supporting evidence, and separately evaluates the retention check.

The recorded human review covers 40 action labels and 38 assessable retention labels, with complete agreement against the judge (Cohen's $\kappa = 1.00$) [2]. The judge and agent share a model, so correlated errors remain possible; LLM-judge biases are discussed by Zheng et al. [9]. Action labels can be ambiguous when an answer mixes mechanisms. In `cs-04a` with write-time Supermemory the answer calls `tenacity` but sets a single attempt, which arguably implements the fail-fast instruction; the judge scored it as the standing retry rule. We did not re-score such cases.

We report 95% Wilson confidence intervals [7] and exploratory exact McNemar tests [4] on the same episodes. Because the A/B probes within a pair share history, episode-level tests do not fully account for dependence; p-values are interpreted cautiously.

## 5 Results and Analysis

All 320 runs completed without execution errors.

### 5.1 Overall Correction Adherence

Figure 3 shows adherence across the eight conditions. No memory achieves 20% (8/40), versus 80% (32/40) with the full transcript. Mem0 and Supermemory achieve 77.5% (31/40) and 85% (34/40).

Read-time tagging achieves 80% with Mem0 and 92.5% (37/40) with Supermemory, the highest observed score. Write-time tagging scores 60% and 77.5%, respectively. Supermemory + read-time resolves 17/20 complete pairs, compared with 15/20 for its baseline.

Most of these differences are within sampling noise (Table 5). Read-time Supermemory's gain over its baseline (3 episodes gained, none lost, $p = 0.25$) has overlapping Wilson intervals (80–97% vs 71–93%). Write-time tagging's drop against the baselines is not significant for Mem0 ($p = 0.118$) or Supermemory ($p = 0.549$). The one significant adherence difference between tagging variants is write-time against read-time Mem0 (2 episodes vs 10, $p = 0.039$). Nothing is corrected for multiple comparisons.

Table 5: Exact McNemar tests on the same 40 episodes ("Only" = episodes where only that condition was right). Retention uses the episodes the judge could assess.

| Metric | Condition | vs | Only cond. | Only base | $p$ |
|---|---|---|---|---|---|
| Adherence | Mem0 + write | Mem0 | 4 | 11 | 0.118 |
| Adherence | Mem0 + read | Mem0 | 2 | 1 | 1.000 |
| Adherence | Mem0 + write | Mem0 + read | 2 | 10 | 0.039 |
| Adherence | Supermemory + write | Supermemory | 4 | 7 | 0.549 |
| Adherence | Supermemory + read | Supermemory | 3 | 0 | 0.250 |
| Adherence | Supermemory + write | Supermemory + read | 3 | 9 | 0.146 |
| Adherence | Mem0 | Full transcript | 1 | 2 | 1.000 |
| Adherence | Supermemory | Full transcript | 3 | 1 | 0.625 |
| Retention | Mem0 | Full transcript | 1 | 12 | 0.003 |
| Retention | Supermemory | Full transcript | 1 | 0 | 1.000 |

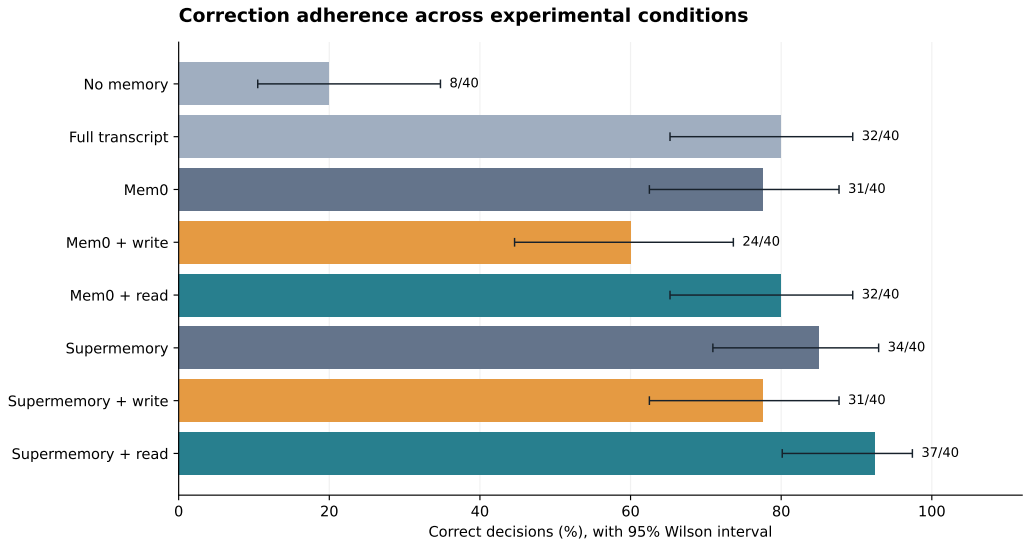



Figure 3: Correction adherence across eight conditions (40 episodes per condition); error bars are 95% Wilson intervals.

## 5.2 Retrieval Versus Application

Across the six memory-system conditions, 43 of 51 incorrect responses occurred despite retrieval of the expected memory (Figure 4). Unmodified Mem0 and Supermemory retrieved the expected memory in all 40 episodes but made nine and six incorrect decisions, respectively. Read-time tagging reduced these counts to eight and three.

Thus, successful retrieval is insufficient for correct application, although a retrieved note may still have lost important scope information.

The cleanest evidence comes from the full-transcript control, where retrieval is perfect by construction yet 8 of 40 answers are wrong. All four single-task B probes (`st-01b` to `st-04b`) fail even though each probe signals a different task ("New task", "Another task", "Different job"). In three of them the agent carried the one-off exception into the new task (CSV, cups, views); in the fourth, `st-03b`, it chose `matplotlib` instead of the project's `plotly`. Retrieval is also an easy step here: each pair has only 3.3 reference memories on average (66 over 20 pairs) and ten results are returned, so near-complete retrieval is expected. Table 6 gives the full breakdown.

Table 6: Retrieved the expected memory vs acted correctly (episodes out of 40). No memory counts as not retrieved; the full transcript as retrieved.

| Condition | Right/right | Right/wrong | Wrong/right | Wrong/wrong |
|---|---|---|---|---|
| No memory | 0 | 0 | 8 | 32 |
| Full transcript | 32 | 8 | 0 | 0 |
| Mem0 | 31 | 9 | 0 | 0 |
| Mem0 + write | 23 | 9 | 1 | 7 |
| Mem0 + read | 32 | 8 | 0 | 0 |
| Supermemory | 34 | 6 | 0 | 0 |
| Supermemory + write | 31 | 8 | 0 | 1 |
| Supermemory + read | 37 | 3 | 0 | 0 |

Columns give retrieval/action ("right/wrong" = retrieved right, acted wrong).

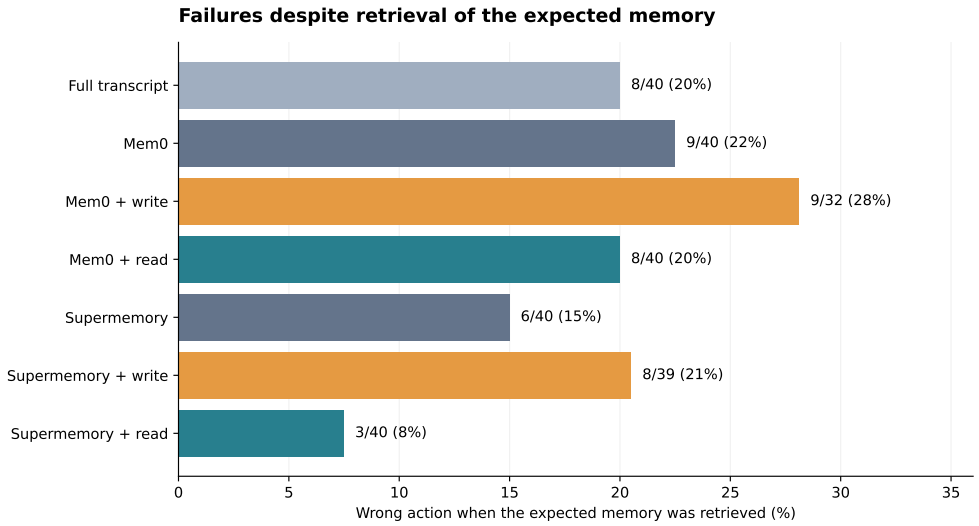



Figure 4: Share of episodes with a wrong action when the expected memory was retrieved (counts over the episodes where it was retrieved). The full transcript counts as retrieved.

### 5.3 Scope-Boundary and Conflict-Type Performance

Figure 5 reveals asymmetric write-time errors. Mem0 + write correctly handles 10/20 in-scope probes, versus 17/20 for Mem0. Supermemory + write handles 11/20, versus 18/20 for its baseline, but correctly handles all 20 out-of-scope probes. This indicates substantial under-application of scoped instructions.

Supermemory + read-time handles 20/20 in-scope and 17/20 out-of-scope probes.

Table 7 shows the direction of the errors. The baselines lean toward over-application (Mem0 5 over-applied vs 4 under-applied; Supermemory 4 vs 1). Write-time tagging shifts errors toward under-application (Mem0: 9 under-applied vs 4 over-applied; Supermemory: 8 vs 0), and Supermemory + read-time makes no under-application errors.

Table 7: Direction of wrong answers (episodes out of 40). Over-applied: a memory outside its scope was used. Under-applied: a broader or less specific memory was used.

| Condition | Over | Under | Superseded | Hedged/other |
|---|---|---|---|---|
| Full transcript | 4 | 3 | 0 | 1 |
| Mem0 | 5 | 4 | 0 | 0 |
| Mem0 + write | 4 | 9 | 1 | 2 |
| Mem0 + read | 4 | 4 | 0 | 0 |
| Supermemory | 4 | 1 | 0 | 1 |
| Supermemory + write | 0 | 8 | 0 | 1 |
| Supermemory + read | 3 | 0 | 0 | 0 |

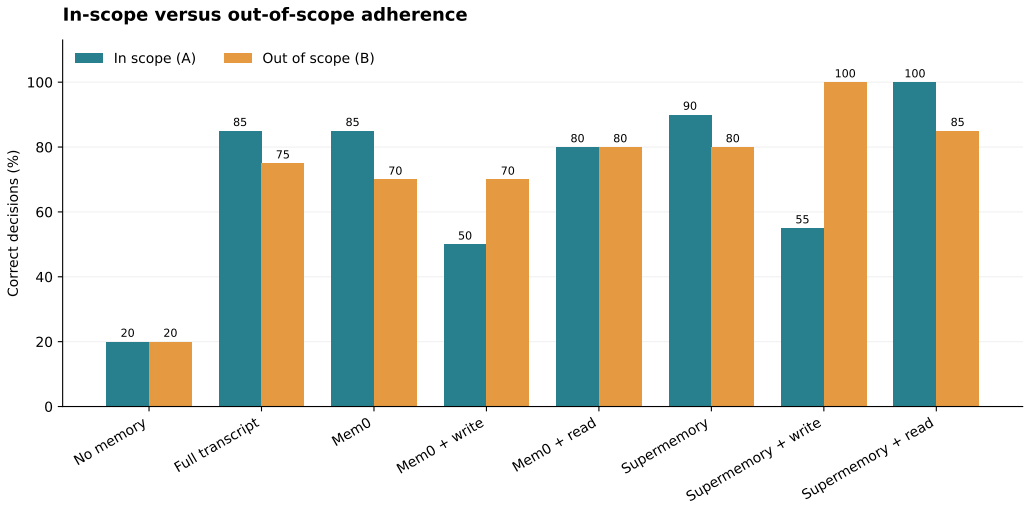



Figure 5: Adherence on in-scope (A) and out-of-scope (B) probes, 20 of each per condition.

Across conflict types (Figure 6), conditional scope and same-scope supersession are handled reliably by most memory-system conditions. Single-task overrides are hardest: Mem0 and Supermemory each achieve 4/8, while read-time Supermemory reaches 6/8. Same-scope specificity also remains challenging, with 4–7/8 correct among the full-transcript and memory conditions.

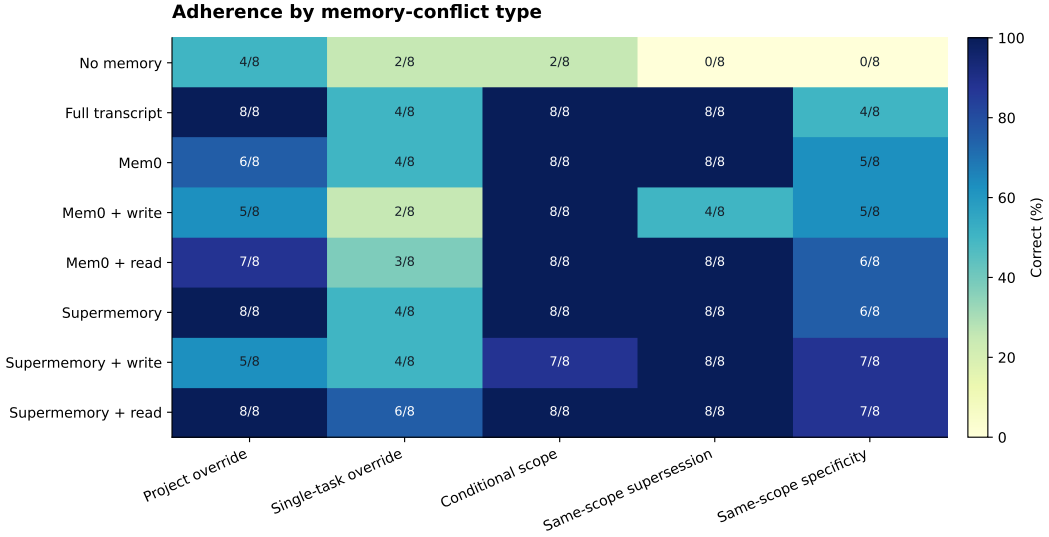



Figure 6: Adherence across five conflict categories; eight episodes per category.

## 5.4 Failure Diagnosis, Tagging, and Retention

The diagnostic breakdown in Figure 7 attributes most Supermemory baseline errors to reasoning despite available information. Mem0 exhibits five baseline scope-loss errors; write-time Mem0 has seven extraction and four tagging failures among its 16 incorrect responses. Supermemory + write-time has eight reasoning-classified failures among nine.

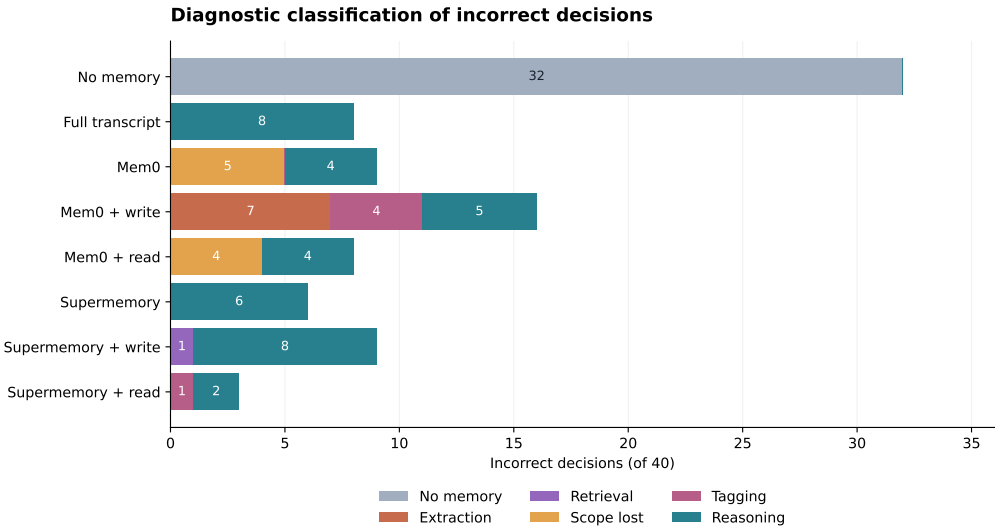



Figure 7: Incorrect decisions by earliest detected failure stage; classifications are heuristic.

Write-time tagging achieves 100% scope accuracy on the tags as the tagger produced them for both systems (124 tags each), versus 87% for read-time Mem0 (111 tags) and 77% for read-time Supermemory (52 tags). For Mem0, only 97 of the 124 write-time tags came back after storage (also 100% accurate), consistent with its re-extraction of the stored instructions. Read-time Supermemory mislabels project memories as standing six times, the commonest confusion there. Nevertheless, the highest downstream adherence occurs with read-time Supermemory, illustrating that tag accuracy alone is insufficient. These figures use different eligible-note sets and are not directly controlled comparisons.

Unrelated-memory retention is 95% with the full transcript, 67.5% with Mem0, and 97.5% with Supermemory. Scope-aware Mem0 variants retain 62–72%, compared with 87–98% for Supermemory variants. Mem0's retention is significantly below the full transcript's (1 episode vs 12, $p = 0.003$), the largest paired difference between a memory system and the control. Scope tagging does not change retention significantly for either system (all $p > 0.2$). Read-time tagging left retention about level (Mem0 66.7%→71.8% on the 39 assessable episodes; Supermemory 97.5%→97.5%), while write-time tagging scored lower in both (Mem0 67.5%→62.5%; Supermemory 97.4%→87.2% on the 39 assessable episodes).

Table 8: Mean OpenRouter cost per episode. Supermemory's own processing is billed separately and is not included.

| Condition | USD / episode | Prompt tokens |
|---|---|---|
| No memory | 0.00012 | 161 |
| Full transcript | 0.00013 | 465 |
| Mem0 | 0.00075 | 19,862 |
| Mem0 + write | 0.00102 | 33,799 |
| Mem0 + read | 0.00093 | 21,954 |
| Supermemory | 0.00012 | 444 |
| Supermemory + write | 0.00034 | 4,571 |
| Supermemory + read | 0.00024 | 2,597 |

Mem0's internal extraction calls make its baseline about 6 times as expensive per episode as the Supermemory baseline on OpenRouter, and its scope-tagged variants 3–4 times as expensive as their Supermemory counterparts. Supermemory's own billing is excluded from this comparison.

# 6 Discussion and Limitations

## 6.1 Write-Time versus Read-Time Tagging

Read-time tagging matches or exceeds adherence relative to both memory-system baselines, whereas write-time tagging scored lower in both (not significantly so against either baseline). For Mem0 the logs point to a concrete cause: each pre-extracted, pre-tagged instruction is passed through Mem0's own extraction again, so 27 of 124 tags were lost (97 returned), 7 of 16 wrong answers are classified as extraction failures, and prompt tokens rise from about 19.9k to 33.8k per episode. This is a property of our integration rather than of write-time tagging itself; storing the tagged instruction without re-extraction would test it.

For Supermemory all 124 tags survive storage, and 8 of its 9 wrong answers are classified as reasoning; the ninth (`st-01a`) is a retrieval miss, where the CSV instruction was not among the returned notes. One possible explanation, which we did not test, is that write-time extraction separates instructions from the conversation that made their scope clear, making task continuity harder to infer. Write-time Supermemory illustrates the trade-off: it avoids over-applying scoped memories on all 20 out-of-scope probes but fails to apply many scoped instructions when appropriate (8 under-applications).

Read-time tagging with Supermemory also works at a coarser grain than Mem0's: Supermemory returns whole session passages, so one tag describes everything in a passage, which may help explain its lower tag accuracy (77%). That coarseness does not explain every error. In `sp-02b` the passage was tagged with the correct condition (`recipient == neighbour`), yet the agent reasoned that a landlord counts as a neighbour (Appendix A.3).

## 6.2 Why Retrieval and Tagging Are Insufficient

The retrieval–application gap shows that access to a relevant instruction does not ensure correct conflict resolution. In pair `po-01`, for example, an agent may select the standing Polars preference despite the applicable project-specific pandas instruction. Other failures involve carrying a single-task exception into a new task (the full-transcript control does this on three of the four single-task B probes and misses the fourth) or combining conflicting instructions.

These cases suggest that effective memory requires both relevant information and reliable contextual decision-making. Representative examples are included in Appendix A.

## 6.3 Threats to Validity

The benchmark contains only 20 synthetic pairs with short histories and explicit scope cues. Nineteen pairs were initially AI-assisted, and numeric conditions avoid threshold boundaries. The findings may not generalize to longer, ambiguous, or naturally occurring interactions. Histories are short enough that retrieval is an easy step (3.3 reference memories per pair against ten results returned), so the retrieval–application gap is measured under favourable retrieval conditions.

The study uses one agent model and one run per condition. Fixed sampling settings do not guarantee reproducibility, while Supermemory's hosted processing is not fully observable.

The scope treatments jointly change metadata, resolution instructions, and memory ordering, preventing attribution of observed effects to scope tags alone. Retrieval matching and failure-stage diagnoses were not independently validated in full. The gold labels were reviewed against the drafts, but the committed annotation files are identical, so no independent inter-annotator agreement is available.

The vendor claim check planned in the project design (a LongMemEval subset run through MemoryBench) was not carried out, so this report makes no comparison with vendors' published figures.

Finally, the two probes within each pair are dependent, limiting the interpretation of episode-level significance tests.

# 7 Conclusion

We evaluated whether scope-aware memory handling helps agents apply retrieved instructions in the correct context. Across 320 runs, incorrect memory application frequently occurred even when the expected memory was retrieved.

Read-time tagging with Supermemory achieved the highest observed adherence (not significantly above its baseline), while write-time tagging scored lower for both memory systems (neither drop against its baseline is significant). Read-time tagging did not lower unrelated-memory retention; write-time tagging scored lower on it, though not significantly. The results suggest that scope-aware interventions may be useful, but explicit metadata alone does not resolve contextual memory conflicts. Larger benchmarks and controlled ablations are needed to isolate the effects of tagging, conflict-resolution instructions, and memory ordering.

# Individual Contributions

| Member | Contributions | Share |
|---|---|---|
| Isaac Chin Wei En | Environment and baseline integration; episode schema, scope-labelling guide and 20-pair benchmark; automated validation and leakage checks; agent-model selection pilot | – |
| Edward Chew Kian Boon | Scope layer and logging | – |
| Ng Cheng Kiat Nicholas | Evaluation and scoring | – |
| Jagan Mohan Shailesh | Validation, analysis, report | – |

# References


[1] Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and D. Yadav. Mem0: Building production-ready AI agents with scalable long-term memory, 2025. URL `https://doi.org/10.48550/arXiv.2504.19413`.

[2] Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960. doi: 10.1177/001316446002000104.

[3] Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. arXiv preprint arXiv:2402.17753, 2024. URL `https://arxiv.org/abs/2402.17753`.

[4] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. doi: 10.1007/BF02295996.

[5] Egor Pakhomov, Erik Nijkamp, and Caiming Xiong. ConvoMem benchmark: Why your first 150 conversations don't need RAG. arXiv preprint arXiv:2511.10523, 2025. URL `https://arxiv.org/abs/2511.10523`.

[6] Supermemory. Research. Supermemory website. URL `https://supermemory.ai/research/`. Accessed 2026-10-26.

[7] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.

[8] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations*, 2025. URL `https://arxiv.org/abs/2410.10813`.

[9] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL `https://arxiv.org/abs/2306.05685`.

# A Episode Examples and Qualitative Failure Analysis

## A.1 Episode JSON Structure

Each benchmark pair is stored as a JSON file under `episodes/pairs/`. The file contains the shared conversation history, reference memories, two evaluation probes, and an unrelated-memory retention check. Table 9 summarises the top-level fields.

Table 9: Top-level fields of an episode-pair JSON file.

| Field | Description |
|---|---|
| `pair_id` | Unique identifier of the paired scenario. |
| `conflict_type` | Type of memory conflict being evaluated. |
| `domain` | Specific decision domain, such as dataframe library. |
| `domain_kind` | Broader category of the task domain. |
| `sessions` | Ordered earlier conversations containing user instructions and contextual information. |
| `memories` | Reference instructions extracted from the sessions, including their source, role, intended action, and gold scope annotations. |
| `probes` | Two evaluation tasks (A and B) sharing the same history but differing in contextual applicability. Each specifies its expected action and winning memory. |
| `retention_checks` | Requirements testing whether unrelated standing preferences are preserved. |

Each session contains a `session_id`, chronological `order`, `task_id`, `active_project`, and conversation `text`.

Each reference memory contains a `memory_id`, source `session_id`, instruction `text`, `role`, `decision`, `action`, and a `gold` object. The latter specifies `scope`, `key`, and `condition`.

Each probe contains a task identifier, project context, natural-language request, structured `facts`, an `in_scope` indicator, the decision being tested, `expected_action`, `expected_memory`, and `deciding_rule`.

The `retention_checks` array identifies an unrelated memory and the requirement used to evaluate whether it is followed.

Only the conversation sessions and the selected probe text are supplied as episode inputs to the experimental systems. Reference memories, structured context, gold labels, and expected outcomes are reserved for validation and scoring.

## A.2 Complete Example: Project Override (`po-01`)

Pair `po-01` illustrates the mirrored-pair design. Two earlier sessions establish three reference memories: a standing preference for Polars on datasets exceeding one million rows, a project-specific instruction to use pandas for Project Marlin, and an unrelated requirement for Python type hints.

Probe A concerns Project Marlin, where the project-specific instruction takes precedence. Probe B changes only the project to Project Tern, where the standing Polars preference applies. Both probes independently test retention of the type-hint instruction.

Listing 1 reproduces the complete JSON specification used by the benchmark.


```
{
  "pair_id": "po-01",
  "conflict_type": "project_override",
  "domain": "dataframe library",
  "domain_kind": "developer",
  "sessions": [
    {
      "session_id": "s1",
      "order": 1,
      "task_id": "t-po01-1",
      "active_project": "Project Marlin",
      "text": "User: I'm kicking off Project Marlin, a nightly sales rollup for the finance team. For Project Marlin, use pandas. My teammate only knows pandas.\nAssistant: Understood, pandas for Marlin. What does the first job look like?\nUser: It reads a CSV with an order_date column written day/month/year. How should I parse those dates?\nAssistant: Tell the parser the day comes first, for example with dayfirst=True, and then check a few rows around the 12th of each month.\nUser: That works. I'll come back with the next job."
    },
    {
      "session_id": "s2",
      "order": 2,
      "task_id": "t-po01-2",
      "active_project": null,
      "text": "User: Our load script reads a big CSV and takes ages. Any quick wins?\nAssistant: Read only the columns you need and give the parser explicit column types. Here is a sketch: def load(path): return read_csv(path, usecols=['id', 'total'])\nUser: That snippet had no type hints. Whenever you write a function for me, give every parameter and the return value a type hint.\nAssistant: Sorry about that. From here on every function I write will have type hints on its parameters and its return value.\nUser: On the library question, I usually go with Polars when a dataset is over 1M rows.\nAssistant: Got it. I'll keep that in mind for large datasets.\nUser: Thanks, that's all for now."
    }
  ],
  "memories": [
    {
      "memory_id": "m1",
      "session_id": "s2",
      "text": "I usually go with Polars when a dataset is over 1M rows.",
      "role": "standing",
      "decision": "dataframe_library",
      "action": "polars",
      "gold": {
        "scope": "standing",
        "key": null,
        "condition": "rows > 1000000"
      }
    },
    {
      "memory_id": "m2",
      "session_id": "s1",
      "text": "For Project Marlin, use pandas.",
      "role": "scoped",
      "decision": "dataframe_library",
      "action": "pandas",
      "gold": {
        "scope": "project",
        "key": "Project Marlin",
        "condition": null
```

```
      }
    },
    {
      "memory_id": "m3",
      "session_id": "s2",
      "text": "Whenever you write a function for me, give every parameter and the return value a type hint.",
      "role": "unrelated",
      "decision": null,
      "action": null,
      "gold": {
        "scope": "standing",
        "key": null,
        "condition": null
      }
    }
  ],
  "probes": {
    "A": {
      "task_id": "t-po01-3",
      "active_project": "Project Marlin",
      "text": "Write a Python function that deduplicates the 5M-row orders table for Project Marlin by order_id, keeping the row with the latest updated_at, and returns the cleaned dataframe.",
      "facts": {
        "rows": {
          "value": 5000000,
          "text": "5M-row"
        }
      },
      "in_scope": true,
      "decision": "dataframe_library",
      "expected_action": "pandas",
      "expected_memory": "m2",
      "deciding_rule": 2
    },
    "B": {
      "task_id": "t-po01-4",
      "active_project": "Project Tern",
      "text": "Write a Python function that deduplicates the 5M-row orders table for Project Tern by order_id, keeping the row with the latest updated_at, and returns the cleaned dataframe.",
      "facts": {
        "rows": {
          "value": 5000000,
          "text": "5M-row"
        }
      },
      "in_scope": false,
      "decision": "dataframe_library",
      "expected_action": "polars",
      "expected_memory": "m1",
      "deciding_rule": 1
    }
  },
  "retention_checks": [
    {
      "memory_id": "m3",
      "check": "The function in the agent's code has a type hint on every parameter and on the return value"
    }
  ]
}
```


Listing 1: Complete JSON specification of project-override pair `po-01`.

### A.3 Qualitative Failure Cases

The following cases illustrate distinct memory-application failures observed in the experimental evaluation.

### A.3.1 Project Override: `po-01`

The user generally prefers Polars for datasets exceeding one million rows but requires pandas for Project Marlin.

Probe A concerns Marlin and expects pandas, whereas Probe B concerns Project Tern and expects Polars. The unrelated type-hint preference applies to both.

On Probe A, four conditions chose Polars where pandas was expected: Mem0, both scope-tagged Mem0 variants, and write-time scope-tagged Supermemory. In all four, the Marlin pandas instruction was among the retrieved memories, and the failure was in the agent's reasoning. Plain Mem0 opened its answer with "Use Polars for this 5M-row dataset to ensure high performance and memory efficiency." In both tagged Mem0 variants, the tagger had labelled the Marlin memory as project scope with key Project Marlin and the Polars memory as standing scope, yet the agent still applied the standing rule.

The full-transcript, Supermemory and read-time scope-tagged Supermemory conditions chose pandas on Probe A. All seven conditions with memory chose Polars on Probe B, as expected. Retrieving the applicable instruction therefore did not ensure that the agent applied it over a conflicting standing preference.

### A.3.2 Single-Task Override: `st-01`

The user usually exports tables as parquet, but for one client-churn export asks for CSV because the retention team opens it in Excel.

Probe A continues that job and expects CSV. Probe B begins "New task, client churn export job: ..." and expects the standing parquet preference.

On Probe B the full-transcript control, read-time Mem0 and read-time Supermemory all answered with CSV; in read-time Mem0 the note recording the CSV request reached the agent untagged, and the agent opened with "Format: CSV." The full-transcript control also fails all four single-task B probes, three of them by applying the exception (CSV, cups, views), so the failure is in the agent's handling of task continuity, not in what was retrieved.

### A.3.3 Same-Scope Specificity: `sp-02`

The user normally begins emails with "To whom it may concern," but when writing to a neighbour wants "Hey there,".

Probe A asks for an email to a neighbour and expects the conditional greeting. Probe B asks for an email to a landlord and expects the default.

On Probe A, plain Supermemory produced both greetings ("Hey there, To whom it may concern,"), which the judge labelled mixed. On Probe B, read-time Supermemory returned a passage tagged with the correct condition (`recipient == neighbour`), yet the agent wrote that it "applied the neighbour-specific opening rule ... as the landlord is a person in the user's building." The failure-stage classifier labelled this case a tagging error; on the evidence above it is better described as the agent stretching the condition.

These examples complement the aggregate results in Section 5. The retrieved notes, responses and judge labels for a representative sample are in `reports/examples.md`; the full per-run logs are not committed (Appendix C).

## A.4 Agent Prompt and a Rendered Example

The same prompt template is used in every condition; only the {`memories`} text changes.

```
You are an assistant helping a user with data and software
work. Below are memories saved from the user's earlier
conversations, followed by their current request. Use the
memories that apply to this request and ignore the ones that
do not.

Memories:
{memories}

Current request:
{probe}

Respond with your answer. State the concrete choice you made
(for example the tool, format or approach) in the first line.
```

For `po-01a` under read-time Mem0, the memories slot held six retrieved notes, for example "User is starting Project Marlin ... and has mandated the use of pandas" tagged *project, key: Project Marlin*, and "User prefers using the Polars library ... larger than 1 million rows" tagged *standing* with condition `rows > 1000000`, followed by the resolution rules of Section 3.1. The agent still answered "I chose Polars ... triggering the standing preference ..."

# B Supplementary Experimental Results

The full pair-by-pair adherence matrix, tagging-accuracy breakdown, failure-stage counts, and per-condition cost accounting are available in `reports/summary.md` and `reports/episode_scores.csv`.

OpenRouter spending for the full experiment was approximately $0.15, with approximately $0.06 for judging. Supermemory's separately billed processing is excluded.

# C Reproducibility and Implementation Details

The implementation is available in the project repository. Experimental conditions are defined in `agent/conditions.py`; runs are executed through `agent/runner.py`; and scoring is implemented in `scoring/`.

The per-run logs (`runs/logs/`) and the judge cache are gitignored and are not in the repository; they were produced on one team member's machine. The committed `reports/` directory holds the scored outputs (`episode_scores.csv`, `tables/`, `examples.md`), so the tables and figures here can be checked from the repository, but re-scoring from raw logs needs those local files. Figures are generated from the tables in `reports/tables/`.